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Influence of Expansion Models and Chamber Pressure
on Ideal Specific Impulse of a LOX/LH2 Rocket Engine

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Abstract

This project presents a Python/Cantera analysis of an idealized liquid oxygen and liquid hydrogen (LOX/LH₂) rocket engine. The study focuses on comparing two fundamental limiting models of nozzle expansion: Frozen Flow and Shifting Equilibrium. The evaluation is conducted across multiple high-performance chamber pressures (50, 100, and 200 bar) specifically to better highlight the differences between these two models and demonstrate how chemical recombination effects scale with pressure. By employing rigorous Gibbs free energy minimization for adiabatic constant-pressure combustion and isentropic nozzle expansion, the results mathematically demonstrate that the shifting equilibrium model predicts a higher specific impulse, and that elevated chamber pressures further enhance this divergence.

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1 Introduction

Liquid oxygen and liquid hydrogen (Hydrolox) form one of the most efficient chemical propellant combinations used in rocketry. The objective of this project is to create a computational tool in Python using the Cantera library to calculate and compare the ideal specific impulse under two limiting assumptions: frozen composition and shifting equilibrium. Additionally, a pressure sweep is conducted up to 200 bar to evaluate the thermodynamic benefit of modern high-pressure combustion cycles comparable to the Space Shuttle Main Engine (RS-25).

2 Theoretical Model and Governing Equations

2.1 Chemical Equilibrium via Gibbs Minimization

To determine the composition of the combustion products in the chamber, Cantera minimizes the total Gibbs free energy of the mixture. For a mixture of K species, the total Gibbs free energy G is:

$$G(T, p, n) = \sum_{i=1}^K \mu_i n_i \quad (1)$$

where n_i is the number of moles of species i , and μ_i is its chemical potential. The minimization is subject to elemental mass conservation constraints.

2.2 Chamber State (HP) and Expansion (SP)

The combustion chamber is modeled as an adiabatic, constant-pressure reactor, imposing the conservation of total enthalpy (H). The nozzle expansion is treated as an ideal isentropic process:

$$s_{exit}(T_e, p_e, X_e) = s_{chamber}(T_c, p_c, X_c) \quad (2)$$

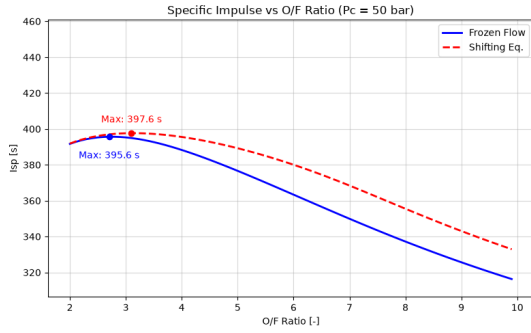
Once the exit state is resolved, the exit velocity (v_e) and specific impulse (I_{sp}) are derived from the macroscopic energy balance:

$$v_e = \sqrt{2(h_c - h_e)}, \quad I_{sp} = \frac{v_e}{g_0} \quad (3)$$

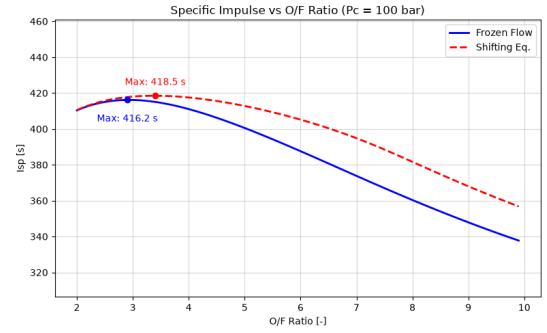
3 Results and Discussion

The automated Python script generated performance curves across the O/F spectrum (from 2.0 to 10.0). The presentation of results for three distinct chamber pressures (50, 100, and 200 bar) was specifically introduced to better highlight the differences and the growing performance gap between the Frozen Flow and Shifting Equilibrium models. To ensure a fair visual comparison across these operating pressures, the Y-axes for all corresponding thermodynamic plots were strictly unified.

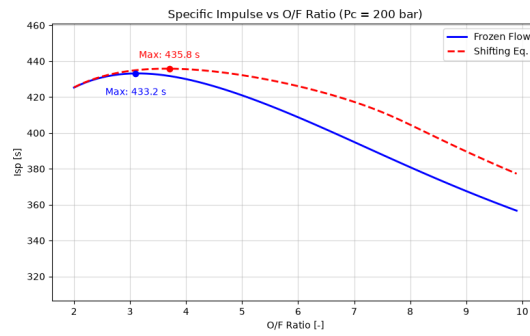
3.1 Specific Impulse Comparison



(a) 50 bar



(b) 100 bar

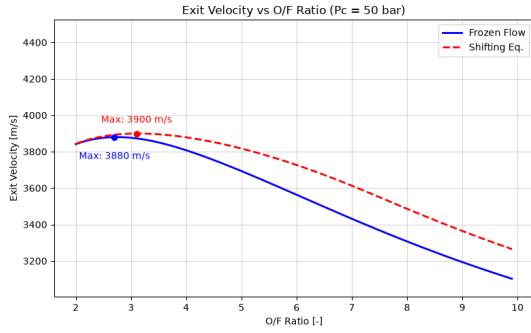


(c) 200 bar

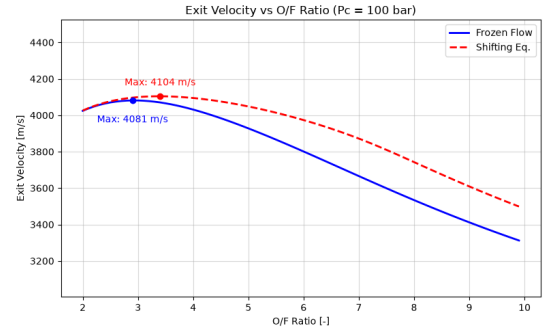
Figure 1: Comparison of Specific Impulse between Frozen Flow and Shifting Equilibrium models.

The shifting equilibrium model consistently yields a higher I_{sp} across all chamber pressures. The multi-pressure layout clearly highlights how the absolute performance difference between the two models becomes more pronounced at higher pressures. This proves the physical reality that chemical recombination in the nozzle is an exothermic process, converting chemical bond energy into kinetic energy.

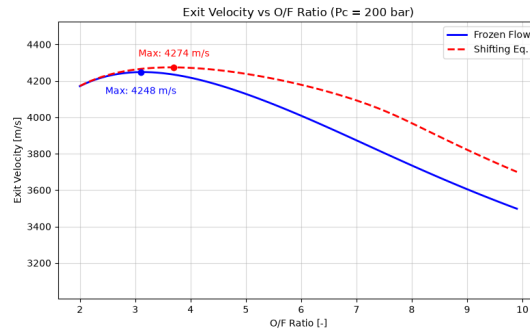
3.2 Exit Velocity Comparison



(a) 50 bar



(b) 100 bar

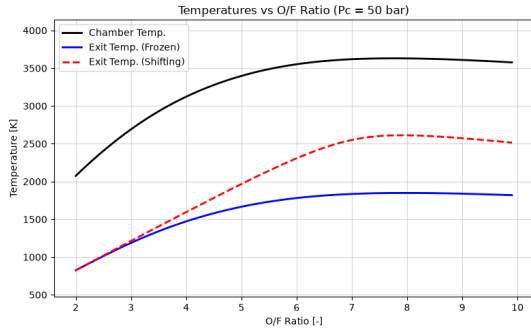


(c) 200 bar

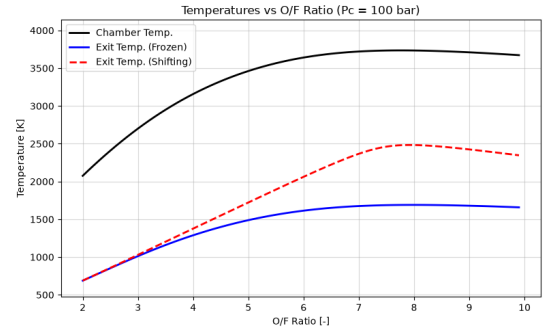
Figure 2: Comparison of absolute Exit Velocity between the two expansion models.

Since specific impulse is directly proportional to exit velocity ($I_{sp} = v_e/g_0$), the exit velocity plot mirrors the I_{sp} trend. The shifting equilibrium model allows the exhaust gases to reach significantly higher velocities, exceeding 4000 m/s for the optimal O/F ratio at higher pressures.

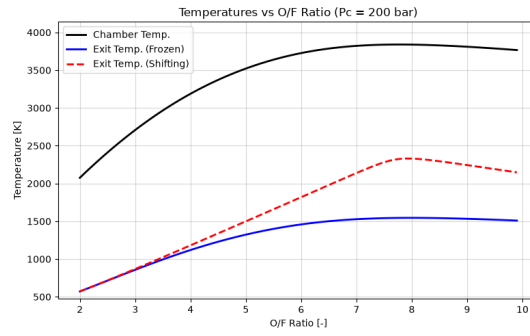
3.3 Thermal Profile in the Engine



(a) 50 bar



(b) 100 bar

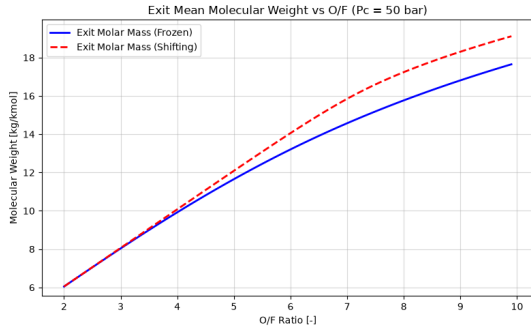


(c) 200 bar

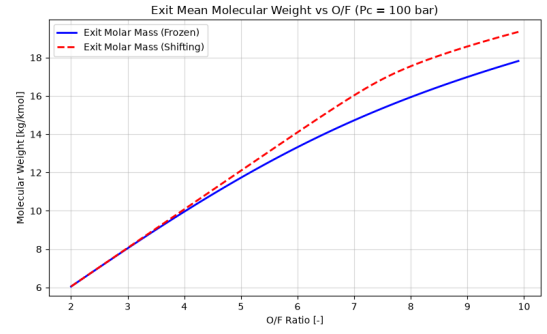
Figure 3: Chamber and exit temperatures across the O/F spectrum.

The exit temperature for the shifting equilibrium model is significantly higher than that of the frozen flow model, as the heat released by recombining radicals actively heats the expanding gas. It can also be observed that increasing chamber pressure slightly raises the peak combustion temperatures.

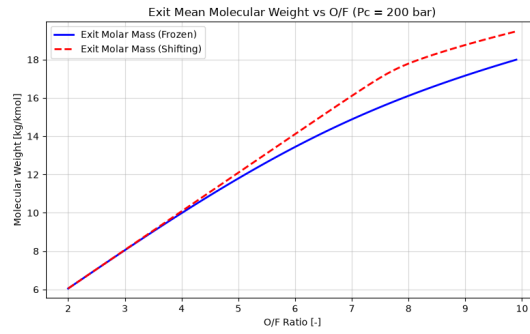
3.4 Exhaust Molar Mass Variation



(a) 50 bar



(b) 100 bar



(c) 200 bar

Figure 4: Mean molecular weight of the exhaust gases at the nozzle exit.

The mean molecular weight of the exhaust in the shifting equilibrium model is higher because light atomic radicals recombine into heavier, stable molecules (H_2O).

3.5 Chemical Recombination Evidence

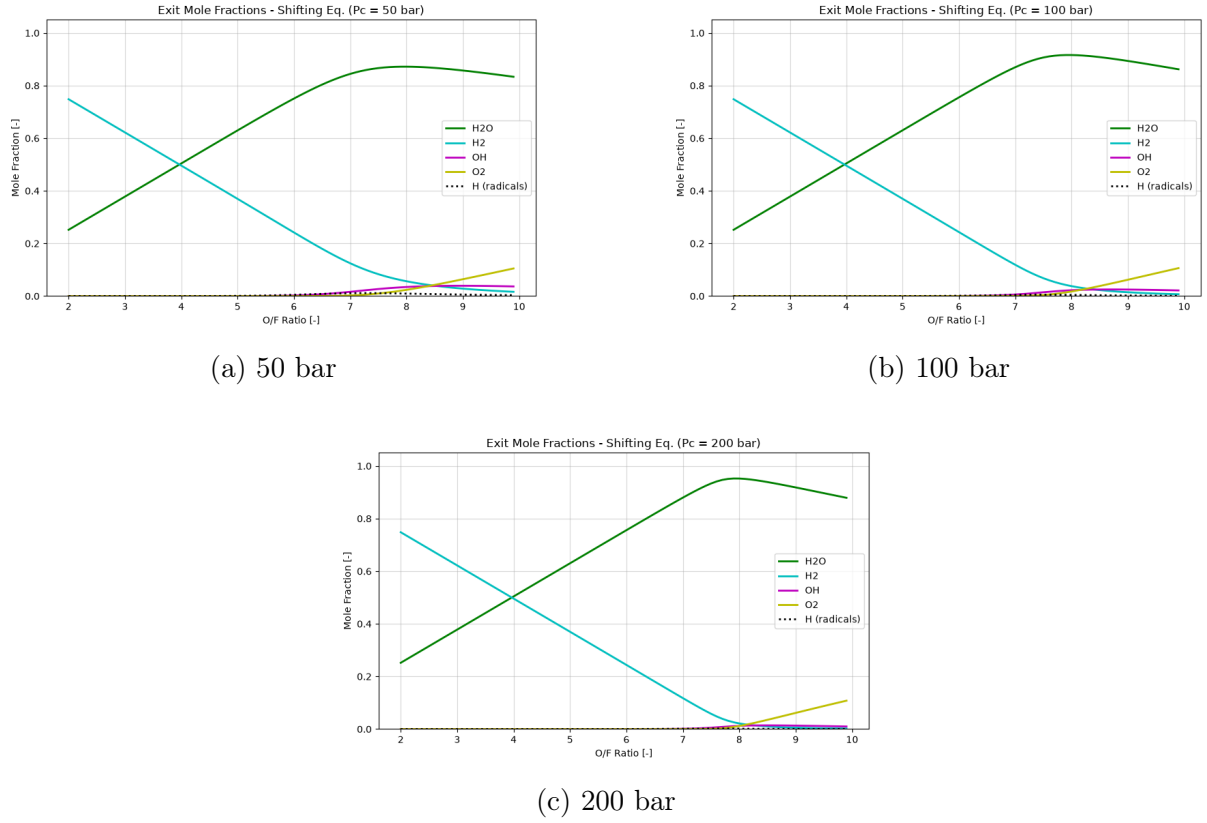


Figure 5: Exit mole fractions of major species and radicals for the Shifting Equilibrium model.

Depending on the O/F ratio, the presence of radicals such as OH and H shifts drastically, showcasing the capability of the Cantera solver to adapt the mixture composition to local isentropic nozzle conditions.

3.6 Influence of High Chamber Pressure

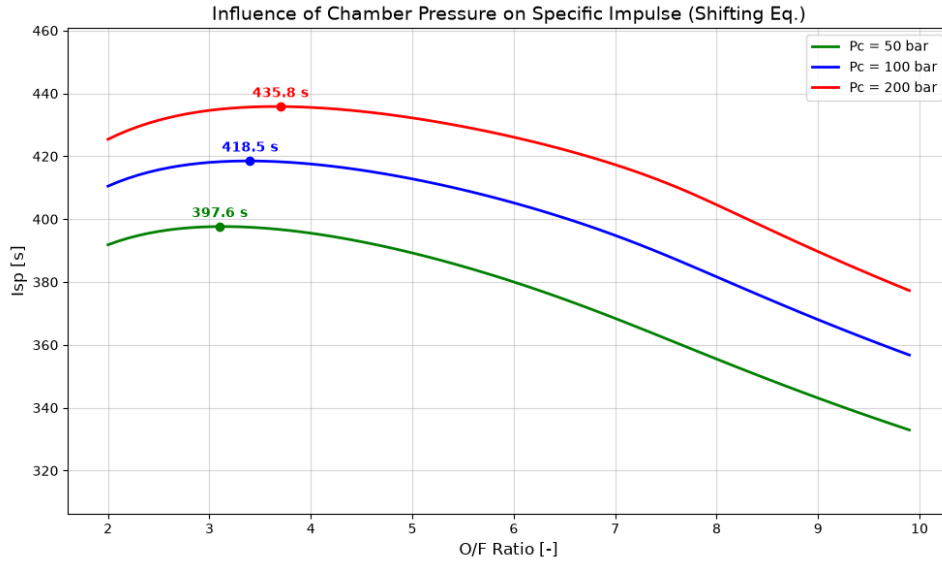


Figure 6: Influence of chamber pressure on the Specific Impulse (Shifting Equilibrium model).

The consolidated pressure sweep clearly indicates that an increase in chamber pressure from 50 to 200 bar provides a steady increase in specific impulse. Comparing the multiple pressure tiers was explicitly chosen to highlight the differences between the models; it vividly illustrates that the thermodynamic benefits of Shifting Equilibrium, driven by a larger available pressure ratio (p_c/p_a) and intense kinetic energy extraction from recombining exhaust gases, are significantly amplified at high pressures.

4 Conclusions

By replacing simplified algebraic formulas with rigorous Gibbs free energy and enthalpy-entropy conservation equations, a highly accurate Python model of a LOX/LH2 engine was constructed. Comparing multiple operating pressures successfully highlighted the stark differences between the expansion models. The shifting equilibrium model proves that recovering dissociation energy provides a tangible performance boost compared to the theoretical frozen limit. Furthermore, the multi-pressure sweep confirms that the gap between these models grows favorably at higher pressures, validating the aerodynamic and thermodynamic benefits of modern high-pressure combustion cycles (up to 200 bar) in liquid rocket propulsion.